SUPPORTING TOOL

Architecture of the interactive simulator

A single bilingual (EN/ES) HTML file that runs the normalized formula in real time: every change recomputes the index and all outputs.

1 · INPUTS

31 sliders

7 at Level 1 (the five components, α and γ) and 24 at Level 2, in four optional blocks: NP, SW, MS and CP can be computed from their sub-components.

→

2 · ENGINE

Normalized formula

```
MP_norm = G^(1−w) · CP^w
```

Weighted geometric mean with $n_{eff} = 3 + \gamma + \alpha$; Proposition 1 live: sign of CP − G.

→

3 · OUTPUTS

- Index, level and CP maturity band
- Gauges and radar by component
- Elasticity, marginal return and cost threshold
- Monte Carlo with 20,000 draws
- Comparison charts

No installation: it opens in any browser and the whole model runs locally.
diegocabuya.github.io/Cyber_Power_and_Maritime_Power/simulator/simulator.html

The 31 controls, component by component

MI	**MI** *Maritime interests*	Direct (weighted sub-index of maritime interests)	**1**
NP	**NP = F × P × MDA$^{\mu}$** *Naval power*	F · P · MDA · μ	**1 + 4**
SW	**SW = (PW·IW·SocW)** *Strategic will*	PW · IW · SocW · w_1 w_2 w_3	**1 + 6**
MS	**MS = (IC·TE·BC)** *Maritime security*	IC · TE · BC · a_1 a_2 a_3	**1 + 6**
CP	**CP = min{1, …}** *Cyber power*	CS_m · CD_m · RC_m · CDip · β_1 β_2 β_3 · δ	**1 + 8**
α γ	**α · γ** *Parameters*	α: weight of cyber dependence · γ: weight of the security environment	**2**

CONTROLS

31

7 at Level 1 + 24 at Level 2

Level-2 values are illustrative and weights are normalized to add up to one.

SIMULATOR · USE

How it works in practice

1

Load a scenario

Colombia (reference case) or the critical, reference and consolidated profiles of the article.

2

Adjust the controls

The index, the sensitivity table and the charts recompute instantly.

3

Read the result

MP_{norm} with its level, the sign of CP − G and the Monte Carlo uncertainty band.

4

Compare with costs

Use the c_k/c_{CP} threshold of Corollary 1 before prioritizing an investment.

PRELOADED SCENARIOS

From the article: Colombia ($MP_{norm} \approx 0.455$) and the critical ≈ 0.19 · reference ≈ 0.52 · consolidated ≈ 0.85 profiles (Table 6).

Ten illustrative scenarios: Brazil, Argentina, US, Canada, Mexico, Chile, Peru, Ecuador, Uruguay and Venezuela, **with a disclaimer**: author-set values, not measurements, not part of the article.

TEACHING VALUE

Driving any component toward zero collapses the index: limited substitutability stops being a theoretical claim and becomes demonstrable live.